

Multivariable Calculus

MATH 1920 Lecture Notes

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1 Parameterization of Surfaces

Problem 1.1: Parameterization of Surfaces

Given $y = \sqrt{x^2 + z^2}$, $x = z^2$, find a parameterization of the surface.

Solution 1.1

Since for any z , $\exists$ a unique position on the curve, we could set $z = t$. Hence,

$$x = t^2$$

$$z = t$$

$$y = \sqrt{t^4 + t^2} = t\sqrt{t^2 + 1}$$

$$\therefore \vec{r}(t) = \langle t^2, |t|\sqrt{t^2 + 1}, t \rangle$$

1.1 Calculus on Vector-Valued Functions

The calculus on vector-valued functions is component-wise.

Generally, for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$, we have

$$\lim_{t \rightarrow t_0} \vec{r}(t) = \left\langle \lim_{t \rightarrow t_0} x(t), \lim_{t \rightarrow t_0} y(t) \right\rangle$$

Similarly, the derivative is defined as

$$\vec{r}'(t) = \langle x'(t), y'(t) \rangle$$

The integral is defined as

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b x(t) dt, \int_a^b y(t) dt \right\rangle$$

Example 1.1 (Calculus on Vector-Valued Functions). Given $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$.

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$$

$$\vec{r}'(0) = \langle 0, 1, 1 \rangle$$

Notice that the result of the derivative is a vector, which is the tangent vector of the curve at $t = 0$.

Now, we want to parameterize the tangent line at $t = 0$. We have

$$\vec{r}(0) = \langle 1, 0, 0 \rangle$$

Hence,

$$\begin{aligned} \ell(t) &= \vec{r}(0) + t\vec{r}'(0) \\ &= \langle 1, 0, 0 \rangle + t\langle 0, 1, 1 \rangle \\ &= \langle 1, t, t \rangle \end{aligned}$$

Problem 1.2: Derivative of Vector-Valued Functions

Find the derivative of the vector-valued function $\vec{r}(t) = \langle t - \sin t, 1 - \cos t \rangle$.

Solution 1.2

The derivative of a vector-valued function is found by differentiating each component.

$$\begin{aligned} \vec{r}'(t) &= \left\langle \frac{d}{dt}(t - \sin t), \frac{d}{dt}(1 - \cos t) \right\rangle \\ &= \langle 1 - \cos t, \sin t \rangle. \end{aligned}$$

We can then compute values of the derivative at specific points.

$$\vec{r}'\left(\frac{\pi}{2}\right) = \left\langle 1 - \cos \frac{\pi}{2}, \sin \frac{\pi}{2} \right\rangle = \langle 1 - 0, 1 \rangle = \langle 1, 1 \rangle$$

$$\vec{r}'(\pi) = \langle 1 - \cos \pi, \sin \pi \rangle = \langle 1 - (-1), 0 \rangle = \langle 2, 0 \rangle$$

$$\vec{r}'(2\pi) = \langle 1 - \cos 2\pi, \sin 2\pi \rangle = \langle 1 - 1, 0 \rangle = \langle 0, 0 \rangle$$

Notice that from the graph, it seems that the curve is “non-differentiable” at $t = 2\pi$. However, the derivative exists at that point, and the tangent vector is $\langle 0, 0 \rangle$, which is a zero vector. This means that the tangent line is not well-defined at that point, but the curve is still differentiable.

Problem 1.3: Implication of $\|\vec{r}'t\|$

If $\|\vec{r}(t)\| = c$ for some constant c , what does this imply about the curve $\vec{r}(t)$?

Solution 1.3

$$\begin{aligned}\|\vec{r}(t)\| = c &\implies \|\vec{r}(t)\|^2 = c^2 \\ &\implies \vec{r}(t) \cdot \vec{r}(t) = c^2 \\ &\implies \frac{d}{dt}(\vec{r}(t) \cdot \vec{r}(t)) = 0 \\ &\implies 2\vec{r}(t) \cdot \vec{r}'(t) = 0 \\ &\implies \vec{r}(t) \cdot \vec{r}'(t) = 0 \\ &\implies \vec{r}(t) \perp \vec{r}'(t)\end{aligned}$$